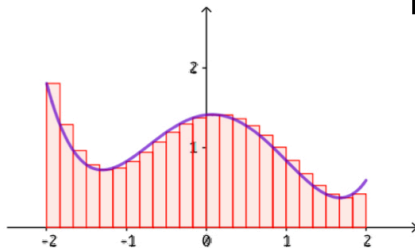


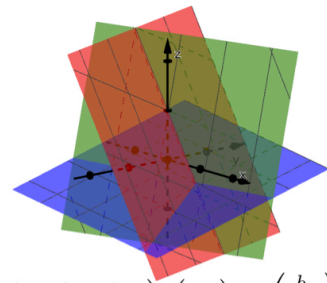
MATH1051

Calculus & Linear Algebra I

Week 10: Vectors & Matrices (IN)



$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x$$



$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

Today's objectives

Part 1 (20 min)

- Introduction & Summary
- Competency questions

Part 2 (30 min)

- Problem Sheet 8: Vectors

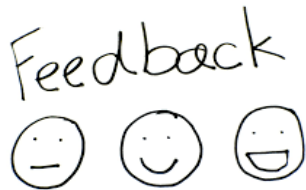
Part 3 (60 min)

- Collaborative Exercise



Week 9 Exercise Feedback

- Marks for graded scripts are available on Blackboard > Gradebook
- Report missing marks using a private post on the Ed Discussion Board
- Common errors/misconceptions.



Vectors Competency Checklist

- V1** I can calculate the magnitude of a vector.
- V2** I can calculate the dot product of two vectors and I understand simple properties of dot products.
- V3** I can determine the angle between two vectors, and determine whether they are parallel or perpendicular (orthogonal).
- V4** I can calculate the projection of one vector onto another, and I understand its geometric meaning.

Matrices Competency Checklist

- M1** I can analyse the sizes of matrices to determine whether they can be multiplied, and determine the size of their products.
- M2** I can perform matrix multiplication.
- M3** I understand the properties of the identity matrix.
- M4** I understand simple properties of matrix transposes. see next slide

Note: Vectors & matrices are revision topics. If you are rusty, there are revision resources on Blackboard.

M4: I understand simple properties of matrix transposes

Note to tutors: Please discuss the following from the workbook:

11.9.1 Examples of transpose

If $A = \begin{pmatrix} 5 & -8 & 1 \\ 4 & 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}$, $C = (7 \ 5 \ -2)$ then find A^T , B^T , C^T .

11.9.2 Properties

For matrices of appropriate size

$$(1) (\alpha A)^T = \alpha \cdot A^T, \quad \alpha \in \mathbb{R}$$

$$(2) (A + B)^T = A^T + B^T$$

$$(3) (A^T)^T = A$$

$$(4) (AB)^T = B^T A^T \quad (\text{not } A^T B^T!).$$

Proofs not required; can easily verify with 2x2 matrices

Vectors

$$\underline{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n, \quad \underline{w} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} \in \mathbb{R}^n$$

Also, two vectors $\vec{u} = \langle u_x, u_y \rangle$ and $\vec{v} = \langle v_x, v_y \rangle$ are parallel to each other if the vector $\vec{u}$ is some multiple of the vector $\vec{v}$. That is, they will be parallel if the vector $\vec{u} = c \vec{v}$, for some real number c . That is, $\vec{u}$ is some multiple of $\vec{v}$.

We define the dot (or scalar) product between $\underline{v}$ and $\underline{w}$ as

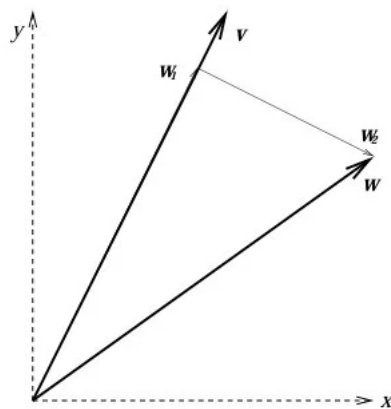
$$\begin{aligned} \underline{v} \cdot \underline{w} &= v_1 w_1 + v_2 w_2 + \cdots + v_n w_n \\ &= \sum_{k=1}^n v_k w_k. \end{aligned}$$

The length or norm of the vector $\underline{v}$ is

$$\begin{aligned} \|\underline{v}\| &= \sqrt{\underline{v} \cdot \underline{v}} \\ &= \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} \end{aligned}$$

(3) If $\underline{v} \neq \underline{0}$, $\underline{w} \neq \underline{0}$, we say that vectors $\underline{v}$, $\underline{w}$ are perpendicular (or orthogonal) if

$$\underline{v} \cdot \underline{w} = 0.$$



We have derived the projection formula:

$$\underline{w}_1 = \frac{(\underline{w} \cdot \underline{v})}{\|\underline{v}\|^2} \underline{v}, \quad \underline{w}_2 = \underline{w} - \frac{(\underline{w} \cdot \underline{v})}{\|\underline{v}\|^2} \underline{v}.$$

↑
The projection of $\underline{w}$ on $\underline{v}$

Past Competency questions

10 mins

1. Let $\mathbf{w} = -3\mathbf{i} + 5\mathbf{j}$. What is $\|\mathbf{w}\|$?
2. For which values of a are the vectors $\mathbf{u} = (a, 4)$ and $\mathbf{v} = (-3, 2)$ parallel?
3. Let $\mathbf{u} = (3, 4)$ and $\mathbf{v} = (-4, 2)$. Find $\mathbf{u} \cdot \mathbf{v}$
4. Let $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$.
 - a) Determine the projection of $\mathbf{v}$ on $\mathbf{w}$
 - b) Determine the projection of $\mathbf{w}$ on $\mathbf{v}$.

For more practice, try the
Vectors competency exercises.

1. Let $\underline{w} = -3\mathbf{i} + 5\mathbf{j}$. What is $\|\underline{w}\|$?

$$\begin{aligned}\underline{w} &= \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} -3 \\ 5 \end{pmatrix} & \|\underline{w}\| &= \sqrt{\underline{w} \cdot \underline{w}} \\ & & &= \sqrt{w_1 \times w_1 + w_2 \times w_2} \\ & & &= \sqrt{(-3) \times (-3) + 5 \times 5} \\ & & &= \sqrt{9 + 25} = \sqrt{34}\end{aligned}$$

2. For which values of a are the vectors $\underline{u} = (a, 4)$ and $\underline{v} = (-3, 2)$ parallel?

Recall: $\underline{u}$ and $\underline{v}$ are parallel if $\underline{u} = b\underline{v}$ for some real number b

$$\text{Let } \underline{u} = b\underline{v} \Leftrightarrow \begin{pmatrix} a \\ 4 \end{pmatrix} = b \begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} -3b \\ 2b \end{pmatrix}$$

$$\Leftrightarrow \begin{cases} a = -3b & (1) \\ 4 = 2b & (2) \end{cases}$$

$$\frac{(1)}{(2)}: \frac{a}{4} = \frac{-3}{2} \Rightarrow a = 4 \times \frac{-3}{2} = -6$$

3. Let $\underline{u} = \begin{pmatrix} u_1 & u_2 \\ 3 & 4 \end{pmatrix}$ and $\underline{v} = \begin{pmatrix} v_1 & v_2 \\ -4 & 2 \end{pmatrix}$. Find $\underline{u} \cdot \underline{v}$

$$\begin{aligned}\underline{u} \cdot \underline{v} &= u_1 v_1 + u_2 v_2 \\ &= 3 \times (-4) + 4 \times 2 \\ &= -12 + 8 = -4\end{aligned}$$

4. Let $\underline{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\underline{w} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$.

We have derived the projection formula:

$$\underline{w} \text{ on } \underline{v}: \quad \underline{w}_1 = \frac{(\underline{w} \cdot \underline{v})}{\|\underline{v}\|^2} \underline{v},$$

a) Determine the projection of $\underline{v}$ on $\underline{w}$

b) Determine the projection of $\underline{w}$ on $\underline{v}$.

$$b) \quad \underline{w}_1 = \frac{(\underline{w} \cdot \underline{v})}{\|\underline{v}\|^2} \cdot \underline{v}$$

$$\underline{w} \cdot \underline{v} = 1 \times 3 + 2 \times 4 = 3 + 8 = 11$$

$$\|\underline{v}\| = \sqrt{1^2 + 2^2} = \sqrt{5} \Rightarrow \|\underline{v}\|^2 = 5$$

$$\underline{w}_1 = \frac{11}{5} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$a) \quad \underline{v}_1 = \frac{(\underline{v} \cdot \underline{w})}{\|\underline{w}\|^2} \cdot \underline{w}$$

$$\underline{v} \cdot \underline{w} = \underline{w} \cdot \underline{v} = 11$$

$$\|\underline{w}\| = \sqrt{3^2 + 4^2} = 5 \Rightarrow \|\underline{w}\|^2 = 25$$

$$\underline{v}_1 = \frac{11}{25} \cdot \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

Part 2: Problem Sheet 8, Vectors

Q1: Use the dot product to determine orthogonality

Q2: Use the dot product to determine the angle between two vectors

Q3: Similar to Q1.

Q4: An introduction to linear combination and consistency of equations.

Q5: Similar to Q2. Algebra for part (b) is tricky

Q6: Projection of a vector on another vector. (similar problem in collaborative exercise).

Q7: Algebra is tricky; Nice intro to cross products

Q8: Application of vectors to velocity problem.

Q9: Application of vectors to 3-d geometry problems

Q10: Dot product wizardry (DIY)

Video solutions available!!! Please refer to these if you need help with questions that are not discussed in class.

Problem Sheet 8: Q4, Q5

Q4, Q5(a) Q9
15 mi

Q4 Let $\mathbf{v} = \begin{pmatrix} 1 \\ -3 \\ 3 \\ 5 \end{pmatrix}$, $\mathbf{w} = \begin{pmatrix} 2 \\ 1 \\ 0 \\ -3 \end{pmatrix}$. Find scalars α, β with $\alpha\mathbf{v} + \beta\mathbf{w} = \begin{pmatrix} 1 \\ -10 \\ 9 \\ 18 \end{pmatrix}$, or show

that no such α, β exist.

DIY, video solution on
Blackboard

Q5 5.

(a) Let $\mathbf{a} = \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} k \\ 2k \\ 3 \end{pmatrix}$. Find all $k \in \mathbb{R}$ such that $\mathbf{a}$ and $\mathbf{b}$ are orthogonal.

(b) For which value(s) of c is the angle between $\mathbf{u} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ c \\ 3 \end{pmatrix}$ equal to 60° ?

(c) Let $A = (4, 2, 0)$, $B = (6, 6, 8)$, $C = (10, 8, 6)$ be points in $\mathbb{R}^3$. Is $\triangle ABC$ right angled?

Problem Sheet 8: Q9

Question **Q9**

- (a) A rectangular box has dimensions $10 \times 15 \times 25$ cm. Position the box along the coordinate axes with the lower left corner at the origin, in such a way that the x , y and z coordinates of all the vertices are ≥ 0 . Draw a picture of this arrangement. What are the coordinates of the 8 vertices?
- (b) Determine the vector $\mathbf{e}$ lying along the shortest edge of the box, and the vector $\mathbf{d}$ joining the origin to the diagonally opposite vertex (located the greatest distance from the origin). What are the lengths of these vectors?
- (c) Determine the angle (to the nearest degree) between $\mathbf{d}$ and $\mathbf{e}$.

Q4. Let $\mathbf{v} = \begin{pmatrix} 1 \\ -3 \\ 3 \\ 5 \end{pmatrix}$, $\mathbf{w} = \begin{pmatrix} 2 \\ 1 \\ 0 \\ -3 \end{pmatrix}$. Find scalars α, β with $\alpha\mathbf{v} + \beta\mathbf{w} = \begin{pmatrix} 1 \\ -10 \\ 9 \\ 18 \end{pmatrix}$, or show

that no such α, β exist.

steps: 1) Find candidate (d, β) ; 2) confirm (d, β)

$$d \cdot \mathbf{v} + \beta \cdot \mathbf{w} = d \cdot \begin{pmatrix} 1 \\ -3 \\ 3 \\ 5 \end{pmatrix} + \beta \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} d \\ -3d \\ 3d \\ 5d \end{pmatrix} + \begin{pmatrix} 2\beta \\ \beta \\ 0 \\ -3\beta \end{pmatrix} = \begin{pmatrix} d+2\beta \\ -3d+\beta \\ 3d \\ 5d-3\beta \end{pmatrix} = \begin{pmatrix} 1 \\ -10 \\ 9 \\ 18 \end{pmatrix} \Leftrightarrow \begin{cases} d+2\beta=1 & (1) \\ -3d+\beta=-10 & (2) \\ 3d=9 & (3) \\ 5d-3\beta=18 & (4) \end{cases}$$

$$\begin{cases} 3d=9 \\ 5d-3\beta=18 \end{cases} \Rightarrow \begin{cases} d=3 \\ \beta=-1 \end{cases} \Rightarrow \text{candidate } (d^*, \beta^*) = (3, -1)$$

$$\text{plug } (3, -1) \text{ into } d \cdot \mathbf{v} + \beta \cdot \mathbf{w} = \begin{pmatrix} d+2\beta \\ -3d+\beta \\ 3d \\ 5d-3\beta \end{pmatrix} = \begin{pmatrix} 3-2 \\ -9-1 \\ 9 \\ 15+3 \end{pmatrix} = \begin{pmatrix} 1 \\ -10 \\ 9 \\ 18 \end{pmatrix}$$

Q5. 5.

(a) Let $\mathbf{a} = \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} k \\ 2k \\ 3 \end{pmatrix}$. Find all $k \in \mathbb{R}$ such that $\mathbf{a}$ and $\mathbf{b}$ are orthogonal.

Recall: $\mathbf{a}$ and $\mathbf{b}$ are orthogonal if $\mathbf{a} \cdot \mathbf{b} = 0$

$$\text{Let } \mathbf{a} \cdot \mathbf{b} = 0$$

$$1 \times k + 4 \times 2k + 3 \times 3 = 0$$

$$k + 8k + 9 = 0$$

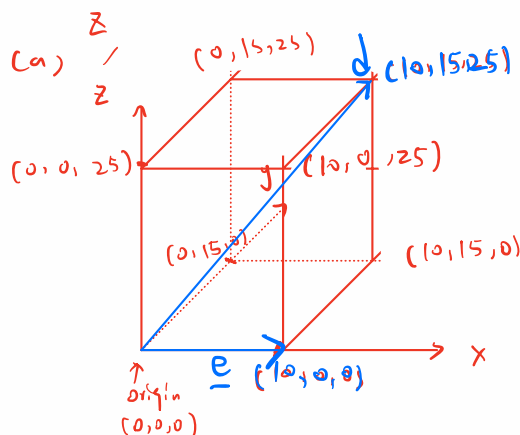
$$k = -1$$

Q6. 9.

(a) A rectangular box has dimensions $10 \times 15 \times 25$ cm. Position the box along the coordinate axes with the lower left corner at the origin, in such a way that the x , y and z coordinates of all the vertices are ≥ 0 . Draw a picture of this arrangement. What are the coordinates of the 8 vertices?

(b) Determine the vector $\mathbf{e}$ lying along the shortest edge of the box, and the vector $\mathbf{d}$ joining the origin to the diagonally opposite vertex (located the greatest distance from the origin). What are the lengths of these vectors?

(c) Determine the angle (to the nearest degree) between $\mathbf{d}$ and $\mathbf{e}$. $\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\| \cdot \|\mathbf{w}\| \cdot \cos \theta$



$$\mathbf{e} = (10, 0, 0) \Rightarrow \|\mathbf{e}\| = \sqrt{10^2 + 0^2 + 0^2} = 10$$

$$\mathbf{d} = (10, 15, 25) \Rightarrow \|\mathbf{d}\| = \sqrt{10^2 + 15^2 + 25^2} = 5\sqrt{38}$$

$$\mathbf{e} \cdot \mathbf{d} = \|\mathbf{e}\| \cdot \|\mathbf{d}\| \cdot \cos \theta$$

where θ is the angle between $\mathbf{e}$ and $\mathbf{d}$

$$\cos \theta = \frac{\mathbf{e} \cdot \mathbf{d}}{\|\mathbf{e}\| \cdot \|\mathbf{d}\|} = \frac{10 \times 10}{10 \times 5\sqrt{38}} = \frac{2}{\sqrt{38}}$$

Part 3: Collaborative Exercise

- **Max group size = 4**
- Discuss the questions with other students at your table.
- Complete the collaborative exercise TOGETHER!
- **Technology free assessment!**
- Your tutors will check your answers and correct any errors. Feel free to move around and use the whiteboard.
- **Submit the collaborative sheet and 'scrap' workings to your tutor.**
- Remember to write your name, ID and AG (Allocated Applied Class Group #) on the hard copy.
- **Note: It is academic misconduct to include names of students who are not present at the practical.**

Hint: Q1. iv. try cross product $\underline{a} = \underline{v} \times \underline{w}$

In **mathematics**, the **cross product** or **vector product** (occasionally **directed area product**, to emphasize its geometric significance) is a **binary operation** on two **vectors** in a three-dimensional **oriented Euclidean vector space** (named here E), and is denoted by the symbol $\times$. Given two **linearly independent vectors** $\mathbf{a}$ and $\mathbf{b}$, the cross product, $\mathbf{a} \times \mathbf{b}$ (read "a cross b"), is a vector that is **perpendicular** to both $\mathbf{a}$ and $\mathbf{b}$,^[1] and thus **normal** to the plane containing them. It has many applications in mathematics, **physics**, **engineering**, and **computer programming**. It should not be confused with the **dot product** (projection product).

$$\mathbf{v} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} d \\ e \\ f \end{pmatrix}$$

$$\mathbf{v} \times \mathbf{w} = \begin{pmatrix} bf - ce \\ cd - af \\ ae - bd \end{pmatrix}$$

11.8 Matrix Multiplication

Q2.

We generalize the multiplication of 2×2 matrices. Let $A = (a_{ij})$ be an $m \times n$ and $B = (b_{jk})$ an $n \times r$ matrix. Then AB is the $m \times r$ matrix with ik entry

$$(ab)_{ik} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{in}b_{nk} :$$

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} b_{11} & \cdots & b_{1k} & \cdots & b_{1r} \\ b_{21} & \cdots & b_{2k} & \cdots & b_{2r} \\ \vdots & & & & \vdots \\ \vdots & & & & \vdots \\ b_{n1} & \cdots & b_{nk} & \cdots & b_{nr} \end{pmatrix} = \begin{matrix} AB \\ (m \times r) \end{matrix}$$

$\begin{matrix} A & B \\ (m \times n) & (n \times r) \end{matrix}$

where $(ab)_{ik}$ = dot product between i th row of A and k th column of B .

AB is only defined if the number of columns of A = the number of rows of B .

Q3.

For matrices, in general, $AB \neq BA$

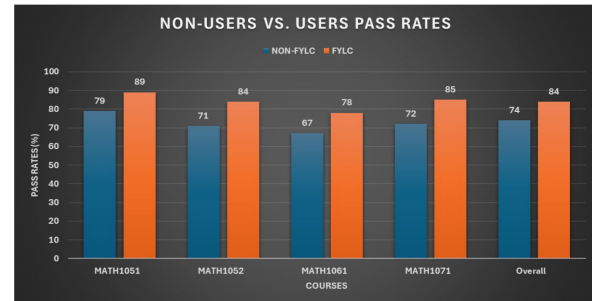
Marking Scheme for Collaborative Exercise

- Mark +0: Your group has no idea!
 - Mark +1: Your group has some understanding but several areas require attention.
 - Mark +2: Your group has a good understanding but some areas need attention.
 - Mark +3: Your group has a very good understanding; minor issues
 - Mark +4: Your group has demonstrated excellent understanding!
-
- Mark +1: Everyone in your group has contributed to completing the collaborative sheet.
Workings on scrap paper will be collected by tutors in addition to the collaborative sheet.
-
- Maximum marks available is 5.

Remember...there is a lot of help available:

Help	Details
Support Learning Tutorial (from Week 8)	Thu 11-12, room 69-304. (note change)
First Year Learning Centre (FYLC)	Mon 11-2, Tue 1-4, Wed 1-4, Thu 11-2, Fri 1-4, 67-443.
Ed online discussion	24/7, staff response within 24 hours (except on weekends!)
Blackboard resources	24/7
Tutors	during applied classes

FYLC, SEMESTER 1, 2024



[More info on Blackboard](#) > [Course Information](#) > [MATH1051 Support](#)

Wrap up! Week 10 → Week 11

By the end of Week 10...

- Complete a selection of exercises from:
 - Competency Exercises – Vectors, Matrices
 - Problem Sheet 8 – Vectors
 - Continue working on A2

Looking ahead to Week 11...

- Topic: gaussian elimination
- All resources for Week 11 are on Bb > Week 11 > Week 11 Applied Classes
- **Assignment 2 due Friday 15th May, 5pm (Week 11)**

Have a great week! See you in Week 11!